

Progress Report 2: Encoder Bandwidth, Not Foveation Specifically, Shapes Spatial Memory in PointNav Agents

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I. QUESTION AND HEADLINE FINDING

A PointNav agent (DD-PPO trained, Habitat sim) has two routes to the spatial information its policy needs: recompute it each step from the current visual encoder output, or maintain it across time inside the recurrent (LSTM) memory. These routes compete for finite hidden-state capacity. *Does the structure of the visual input determine how this capacity gets allocated?* Our claim: a bottleneck at the encoder forces the LSTM to integrate the GPS sensor across time and store the result in a linearly-readable axis of $\mathbf{h}_2$ — it has no other route; a rich encoder offers a viable per-step visual-feature route, and policy gradient reallocates capacity to it. The same hidden state encodes position in both regimes; only the *format* differs.

II. FIVE CONDITIONS ON ONE BANDWIDTH AXIS

All agents share a 3-layer LSTM (512-d) and the Wijmans non-visual sensor stack (goal-in-start-frame, GPS, compass, close-to-goal scalar); only the visual input changes. All four sighted conditions were retrained from scratch under unified hyperparameters (seed=0, num_envs=16, 250M frames), eliminating the inconsistency present in our *Report 1* runs. All achieve 96–99% success and $\text{SPL} \in [0.85, 0.89]$, so any cross-condition representational difference cannot be attributed to a skill gap.

Table I

FIVE CONDITIONS, ORDERED BY ENCODER SPATIAL OUTPUT. BLIND USES AN INDEPENDENTLY-RETRAINED CHECKPOINT AT 340M FRAMES.

Condition	Encoder out	Visual input
Blind	none	no RGB
Coarse	1×1	48×48 RGB downsampled
Foveated-logpolar	2×2	64×64 log-polar resampled
Foveated	4×4	256×256 Gaussian-blurred ($\sigma_{\text{max}}=8$)
Uniform	4×4	256×256 full-resolution RGB

For each condition we collect 500 deterministic-policy episodes and run six analytical lenses on the top-layer hidden state $\mathbf{h}_2$: linear (Ridge $\alpha=10$) and MLP (256-d, $L_2=10^{-4}$) probes for GPS; lag- k probe for $k \in \{0, 1, 2, 5, 10, 20, 50\}$; per-(unit, scene) Skaggs spatial information (rectified $\max(\mathbf{h}_2, 0)$); cross-training probes at five intermediate checkpoints; and cross-condition CKA, Procrustes shape distance, and memory transplant. All probes use 5-fold episode-level cross-validation. Single seed per condition follows the comparative-cognition modelling convention; rigour at the cross-condition level instead derives from convergence across the six lenses.

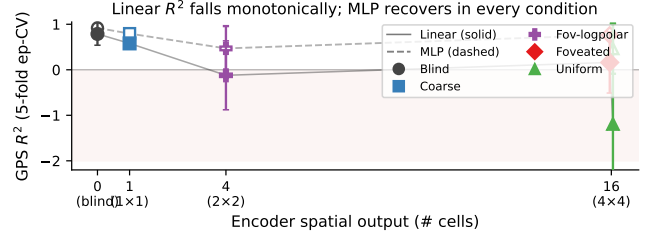


Figure 1. Top-layer GPS R^2 (5-fold ep-CV, mean $\pm \sigma$) vs. encoder spatial output. Solid: linear (Ridge $\alpha=10$) probe. Dashed: 2-layer MLP probe on the same hidden states. Blind point: preliminary 100-ep partial NPZ.

III. RESULTS: A BECAUSE-BUT-THEREFORE CHAIN

Because. Linear GPS R^2 falls monotonically with encoder spatial output (Figure 1, solid line) — a robust trend across the four converged sighted conditions plus the independently-retrained blind. The monotonicity is what made foveated-logpolar (a 2×2 -output rich-encoder control we added *after* Report 1) a *falsification target*: the encoder-bandwidth mechanism predicts it lands in the rich-encoder regime, and Figure 1 shows it does — decisive evidence that the relevant axis is encoder bandwidth, not foveation specifically. The most direct alternative — that the linear probe simply runs out of capacity in rich-encoder hidden states — is ruled out by distance-to-goal, a same-dimensional control target which all five conditions decode at $R^2 > 0.8$.

But. A 2-layer MLP fit on the *same* hidden states recovers GPS in every condition (Figure 1, dashed line), and the gap from linear widens with encoder bandwidth. Position information is not destroyed by a rich encoder; it gets pushed into non-linear $\mathbf{h}_2$ directions the linear DD-PPO action head cannot reach.

Therefore. The encoder and the recurrent state compete for representing position. A bottleneck (no encoder, or 1×1) leaves only one route: integrate the GPS sensor across time and store the result in a linearly-readable axis. A rich encoder opens a per-step visual-feature route; policy gradient reallocates capacity to it, and the integrated code stops being the linearly-readable axis — though, as the MLP recovery shows, position remains encoded. *Format differs; total information shifts only modestly.*

IV. CONVERGENT EVIDENCE

Capacity allocation is a training-time process, not an endpoint property. All four sighted conditions begin training with a high linear top-layer GPS code (Figure 2); the linear axis is the LSTM’s *default* format. What separates conditions is what happens after: bottleneck preserves the integrated code; rich-encoder conditions decay as the visual

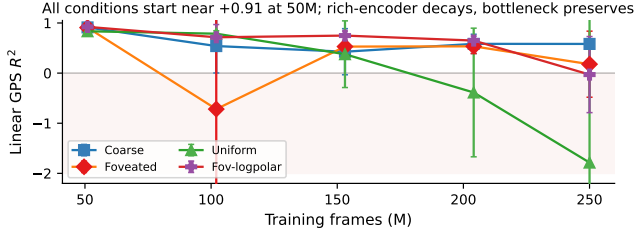


Figure 2. Linear GPS R^2 across training (5 checkpoints per condition; blind has only the 340M ckpt and is omitted). Negative- R^2 outliers clipped to -2.0 .

route consolidates. This rules out the alternative that bottleneck conditions never learned the visual route — they did, but policy gradient kept the integration route on top because there was no usable visual alternative.

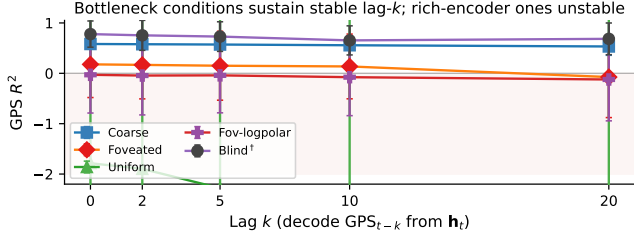


Figure 3. GPS R^2 (5-fold ep-CV) for past-position probes GPS_{t-k} from current $\mathbf{h}_t$. Blind point: preliminary 100-ep partial NPZ.

The bottleneck-state code is integrated; the rich-encoder state is reactive. Decoding past position GPS_{t-k} from current $\mathbf{h}_t$ (Figure 3) shows bottleneck conditions sustain stable positive R^2 across the lag range while rich-encoder ones degrade rapidly. The contrast is about format, not total information: same MLP recovery (Figure 1), different temporal dynamics. Per-(unit, scene) Skaggs spatial information is condition-flat at $[1.16, 1.35]$ bits, within the within-condition spread — ruling out the null reading that bottleneck conditions just have more place cells; the signal lives in the *linear-readout axis*, not the per-unit metric.

V. DEVIATIONS FROM THE PROPOSAL, WITH EXPLANATIONS

H1 reframed. The proposal cast H1 as compensatory memory specifically in the foveated agent (eccentricity-stratified probe accuracy). *Report 1*, on partially-converged checkpoints with heterogeneous hyperparameters, reported foveated as the longest-memory condition. Both the proposal framing and the *Report 1* result are *superseded*. After retraining all four sighted conditions under unified hyperparameters (the original cause of the apparent foveated specialness), and after adding a *foveated-logpolar* control with intermediate (2×2) encoder spatial output as a falsification test, the data show a monotonic encoder-bandwidth-vs-linear- R^2 trend (Figure 1) under which foveated and uniform sit in the same rich-encoder regime. The mechanism is encoder bandwidth, not foveation specifically.

H3 de-scoped, methodology broadened. The proposal’s H3 (epistemic gaze: learned-gaze MLP targeting memory-

weak regions) was de-scoped after the gaze decoder collapsed to a fixed point under pure navigation reward; a bounded- σ stochastic-gaze fix is in the codebase but not pursued this iteration. Compensating, we imported a richer set of cognitive-neuroscience methods into the representational-analysis frame: Skaggs spatial information as a place-cell signature on $\mathbf{h}_2$, an SR-style predictive-horizon probe, Procrustes shape-metric and principal-angle subspace divergence, an excursion-forgetting protocol, and a leave-one-scene-out scene-invariance test. The contribution shifts from a single epistemic-gaze deliverable to recasting classical cognitive-neuroscience metrics as a computational framework for representational analysis of deep-RL agents.

VI. PROBLEMS ENCOUNTERED, SOLUTIONS, AND FORWARD RISKS

Encountered & solved. (i) *Hyperparameter drift across Report 1 conditions* (different num_envs, lr, frame budgets) confounded cross-condition comparisons. Resolved by re-training all four sighted conditions from scratch under a unified config. (ii) *Skaggs spatial information* $\sim 100\times$ inflated relative to the place-cell literature when computed on raw $\mathbf{h}_2$, because near-zero global means dominate the ratio in $I = \sum p(b)(\lambda_b/\bar{\lambda}) \log_2(\lambda_b/\bar{\lambda})$. Resolved by applying ReLU rectification ($\max(\mathbf{h}_2, 0)$) before computing per-bin firing rates, recovering paper-magnitude bits. (iii) *Cross-condition memory transplant on coarse* failed at the state-dict load (incompatible RGB shape: 48×48 vs 256×256). Resolved by restricting to the 6 sighted-only pairs sharing 256×256 input; coarse-and-blind transplants pending an architecture-side fix. (iv) *Blind probe-collect at 50 episodes gave catastrophically wide CV folds* (linear -0.27 ± 1.41). Diagnosed as sample-size noise rather than a methodological bug: the same probe at 100 episodes stabilises to $+0.79 \pm 0.25$.

Likely to encounter. (a) *Blind 500-ep collection ETA* (~ 6 h on RCP) overlaps with the parallel NeurIPS submission deadline; the 100-ep partial NPZ provides a fall-back. (b) *Single-seed per condition* bounds the seed-variance error bars; multi-seed re-runs across all five conditions are out of compute scope for this iteration. (c) *Transplant matrix coverage* is incomplete; without coarse-and-blind pairs we cannot test the bottleneck-vs-rich-encoder asymmetry directly.

VII. ACTION ITEMS TOWARD THE FINAL REPORT

Bandwidth-allocation claim — **A1**: complete blind 500-ep probe-collect, populate blind row across all lenses, tighten Figures 1 and 3; **A2**: retrain coarse with a 256×256 -padded input pipeline so the transplant matrix can include the bottleneck-vs-rich-encoder asymmetry test (closes the H2 format-axis evidence gap); **A3**: two-seed validation on at least the two endpoint conditions (coarse + uniform) to bound seed-variance on the magnitude axis. *Cog-neuro methodology* — **A4**: report the LOSO scene-invariance and excursion-forgetting results in the final write-up (data complete, paper-text deferred); **A5**: place-vs-view-cell dissociation on existing rate-map data; reattempt H3 with the bounded- σ stochastic gaze if time permits.